MATH 231 – Linear Algebra I

The Vector Unit, Part 1 ▪ Vectors: Definition, Notation, and Arithmetic

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About this document. The vector unit is split across three parts, of which this is the first. Section numbering runs continuously through all three, so a reference such as §11.6 always points to the same place regardless of which part you are reading:

Part	Sections	Subject
1	§1–§9	vectors, notation, addition, scaling, equality
2	§10–§12	orientation, the dot product, projection
3	§13–§15	Cauchy–Schwarz, metrics, norms

What you should be able to do afterwards. State what a vector is in terms of magnitude and direction; identify the tail and head of a vector in the plane; compute the Euclidean norm and explain why it is a *unary* operation; distinguish vector quantities from scalar quantities; read all three notational conventions for a vector, and read a signature $f : A \to B$; add and subtract vectors both componentwise and geometrically, using the parallelogram rule and the triangle inequality; scale a vector by a real number and say what $\alpha = -1$ does to it; and decide when two vectors drawn in different places are the *same* vector.

Further reading. Boyd & Vandenberghe, *Introduction to Applied Linear Algebra*, Ch. 1 (vectors, addition, scalar multiplication) and Ch. 3 (norm and distance); Strang, *Linear Algebra and Its Applications*, Ch. 1; Axler, *Linear Algebra Done Right*, Ch. 1.

1. Vectors: magnitude, direction, and the space they live in

We begin with a working definition — deliberately informal, and stated in the language of quantities rather than axioms.

Definition 1.1 (Vector, working definition)**.** A *vector* is a mathematical object carrying two attributes: a **magnitude** and a **direction**.

Looking ahead. This is the physicist's definition, and it is the right one to start from because it is the one you can *see*. Later in the course we replace it with the algebraist's definition: a vector will become any element of a set equipped with two operations obeying a short list of axioms — at which point functions, polynomials, and matrices all become vectors too, and "magnitude and direction" stops being the defining feature. For now, hold on to the picture.

The space a vector lives in

A vector does not exist in isolation; it lives in a space. Ours today is the **two-dimensional Euclidean plane**, the familiar xy-plane, and every vector we draw will live in it.

The word *dimension* here means a **degree of freedom** with respect to spatial coordinates: how many independent numbers must you supply to pin down a location? On a line, one; on the plane, two — an x and a y, chosen independently of one another. The plane is two-dimensional in exactly that sense.

Tail, head, and the two attributes

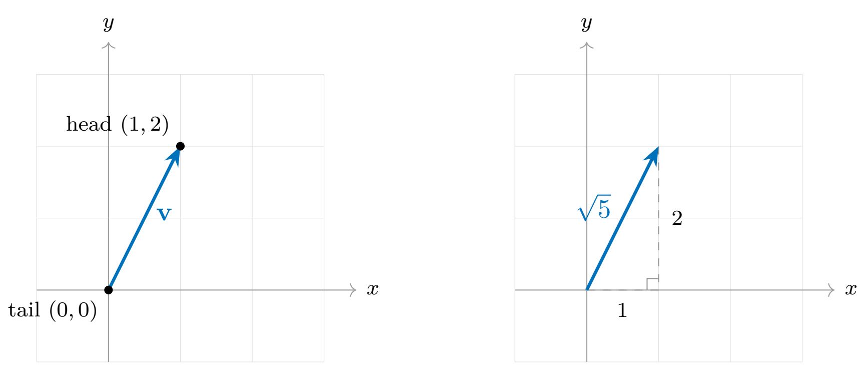
Fix the point $(1, 2)$. Draw an arrow from the origin to it. That arrow is a vector, and it has two distinguished points:

- the **tail** — where the arrow begins, here the origin $(0, 0)$;
- the **head** — where the arrow ends and the arrowhead is drawn, here $(1, 2)$.

With the tail and head named, the two attributes become concrete:

- **Direction** is the arrow's orientation in the plane — which way it points, relative to the axes, as it travels from tail to head. The arrow to $(1, 2)$ points up and to the right, more steeply up than rightward.
- **Magnitude** is the arrow's **length**: the distance from the tail to the head. We measure it with the Euclidean distance, and we compute it in §2.

Note the arrowhead in the left-hand figure below: the arrowhead *is* the direction, so a bare line segment would lose half the information.



2. The norm — our first vector operation

The magnitude of a vector is not just a description of it; it is something we *compute* from it. Computing it is our first vector operation, and it has a name.

Definition 2.1 (Euclidean norm)**.** For a vector $\mathbf{v}$ in the plane with tail at the origin and head at (x, y), the **Euclidean norm** of $\mathbf{v}$ is

$$\|\mathbf{v}\| \;=\; \sqrt{x^2 + y^2},$$

the Euclidean distance from the tail to the head.

Two things to notice about the shape of this operation.

- It is **unary**: it takes *one* vector as input. Compare $3+5$, where addition consumes two numbers — a *binary* operation. The norm consumes one vector and returns one number.
- It turns a *vector* into a *scalar*: the input is a vector in the plane, the output is a single real number, and that number is never negative. Length cannot be negative.

The formula is Pythagoras, nothing more. The right-hand figure above is the proof: the horizontal leg has length x, the vertical leg has length y, and the vector is the hypotenuse.

Example 2.1 (The running vector)**.** For $\mathbf{v}$ with head at $(1, 2)$,

$$\|\mathbf{v}\| \;=\; \sqrt{1^2 + 2^2} \;=\; \sqrt{1+4} \;=\; \sqrt{5} \;\approx\; 2.236.$$

Note that $\sqrt{5}$ is a finished answer: a norm is generically irrational, and there is no need to convert it to a decimal.

Example 2.2 (A clean one)**.** For $\mathbf{w}$ with head at $(3, 4)$, $\|\mathbf{w}\| = \sqrt{9+16} = \sqrt{25} = 5$.

Remark. So three phrases now name the same number: the *magnitude* of the vector, the *length* of the arrow, and the *Euclidean norm* $\|\mathbf{v}\|$. Use them interchangeably.

3. Bridge to calculus: the derivative as a vector quantity

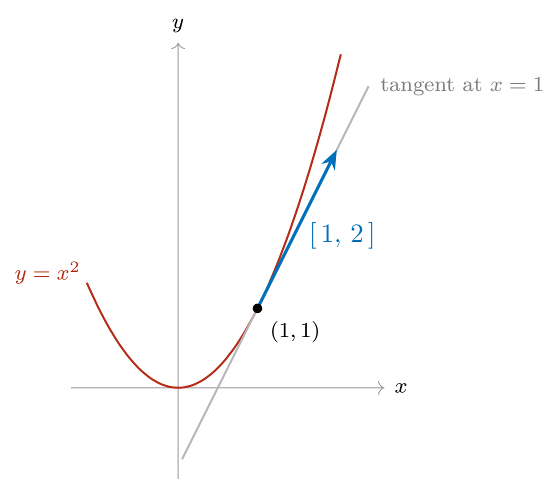
Vectors are not new to you — you have been computing one all through calculus without calling it that.

Let f be differentiable and consider the tangent line to the curve $y = f(x)$ at a point. The tangent line has both of our attributes:

- **Direction.** The *sign* of the slope orients it. A positive slope tilts up-and-to-the-right and the function is increasing there; a negative slope tilts down-and-to-the-right and the function is decreasing. The tangent line points somewhere.
- **Magnitude.** The *size* of the slope measures how fast the function is changing — the “speed” of change in that direction. A slope of 10 climbs ten times as urgently as a slope of 1.

Direction plus magnitude: the derivative of f at a point can be read as a **vector quantity**.

Concretely, take $f(x) = x^2$ at $x = 1$. Then $f'(1) = 2$, so from the point $(1, 1)$ the tangent advances 1 unit rightward for every 2 units upward: its direction vector is $[\,1,\,2\,]$ — the very vector we drew in §1, now carrying a meaning. Its norm, $\sqrt{5}$, is the rate at which arc length accumulates along the tangent per unit of horizontal travel.



4. Vector quantities vs. scalar quantities

The phrase from §3 is worth making official, because it sorts most of the quantities you will meet in the sciences into two bins.

- A **vector quantity** has a magnitude *and* a direction. *Examples:* displacement, velocity, acceleration, force, momentum, torque, the electric and magnetic fields, a temperature *gradient*.
- A **scalar quantity** has a magnitude only — it indicates *scale*, and nothing else. *Examples:* mass, speed, temperature, energy, work, electric charge, time, pressure, density, volume.

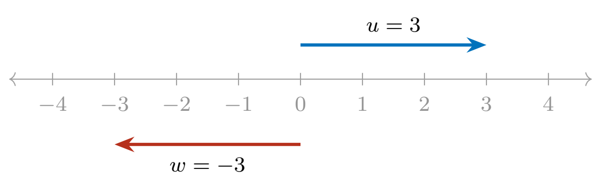
The pair to dwell on is **speed** and **velocity**. Speed is a scalar: 60 mph. Velocity is a vector: 60 mph *due north*. Speed is exactly the magnitude of the velocity vector — that is, its norm — which is why the two words are not synonyms and why a car going round a bend at constant speed is nonetheless accelerating.

5. One-dimensional vectors

Nothing above required two dimensions. In one dimension — on the number line — a vector is a signed number, and the two attributes are still both present:

- the **sign** carries the direction: positive points one way along the line, negative the other;
- the **absolute value** carries the magnitude.

So $u = 3$ and $w = -3$ have the same magnitude and opposite directions. And the one-dimensional norm is $\|u\| = \sqrt{u^2} = |u|$ — absolute value was your first norm all along.



6. Notation, and where vectors sit set-theoretically

There are several standard ways to denote a vector symbolically. Using the letter v purely as an example — *any* letter may be used:

Notation	Reading
$\mathbf{v}$	boldface — common in applied and computational writing
$\vec{v}$	arrow overhead — common in physics and calculus texts
v	plain — common in pure linear algebra, where context suffices

In this class we will use a mixture of all three, and context will tell you which object you are looking at. Getting comfortable reading all three is part of the course: you will meet all of them in the reference texts.

The set-theoretic statement

Until now we have said "the plane" in words. Here is its name. To say that $\mathbf{v}$ is a two-dimensional vector, we write

$$\mathbf{v} \in \mathbb{R}^2, \qquad \text{where} \qquad \mathbb{R}^2 \;=\; \mathbb{R} \times \mathbb{R} \;=\; \big\{(x, y) \;:\; x \in \mathbb{R},\; y \in \mathbb{R}\big\}.$$

Read aloud: "$\mathbf{v}$ is in $\mathbb{R}$-two." The superscript 2 is the dimension of §1 — two coordinates, each free to be any real number, chosen independently. From here on, $\mathbb{R}^2$ replaces "the plane" as our standard name for the space a two-dimensional vector lives in.

The numerical convention for this class

We write a vector in brackets, listing its coordinates in order:

$$\mathbf{v} = [\,x,\; y\,], \qquad \text{for instance} \qquad \mathbf{v} = [\,1,\; 2\,].$$

Order matters: $[1, 2] \neq [2, 1]$.

Naming a function by what it takes and what it returns

A steady stream of operations starts in §7, and we will want a compact way to record what each one consumes and what it produces — before saying a word about how it computes. The standard device is an arrow:

$$f : A \to B,$$

read "f from A to B" or "f maps A to B." It asserts that f accepts an input drawn from the set A and returns an output lying in the set B. The set A is called the **domain**, the set B the **codomain**, and the statement $f : A \to B$ as a whole is the **signature** of f.

Notice what a signature deliberately leaves out: the rule. It is a statement about *types* — what goes in, what comes out — and two operations that compute completely different things may share one.

Two inputs. Some operations take a pair of arguments rather than one, and the Cartesian product above is exactly what is needed to say so. Just as $\mathbb{R}^2 = \mathbb{R}\times\mathbb{R}$ meant "ordered pairs of real numbers,"

$$\mathbb{R}^2 \times \mathbb{R}^2 \quad \text{means} \quad \text{"ordered pairs of vectors."}$$

So an operation swallowing two vectors and returning a vector has signature $\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2$, and one swallowing a number and a vector has signature $\mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}^2$. The number of slots on the left is the number of arguments.

The placeholder dot. An operation written with a symbol rather than a letter needs some way to show its empty argument slots, and the convention is a centred dot, one per slot. The norm of §2, for instance, has signature

$$\|\cdot\| \;:\; \mathbb{R}^2 \to \mathbb{R},$$

where the dot marks the place a vector will go. Read it aloud: "the norm takes a vector in $\mathbb{R}^2$ and returns a real number." That is exactly what §2 said in words — one vector in, one number out — so the notation is not new content, only a shorthand for something you already know.

It also shows the limits of the device. §2 also told us the output is never negative, and the signature cannot say so: its codomain is $\mathbb{R}$, not the non-negative reals. A signature records the shape of an operation, not its finer properties.

Looking ahead. For now we draw every vector with its tail at the origin, so that its head sits at (x, y). That is a convention, not a restriction: in §9 we will see that a vector is really a *displacement*, so the same vector may be drawn with its tail anywhere in the plane. That fact is what makes the head-to-tail picture of §7.1 legitimate.

7. Addition and subtraction

Many operations on vectors await us. Here are the first two.

Definition 7.1 (Addition and subtraction in $\mathbb{R}^2$)**.** Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ with $\mathbf{u} = [x_1, y_1]$ and $\mathbf{v} = [x_2, y_2]$. Then

$$\mathbf{u} + \mathbf{v} = [\,x_1 + x_2,\ \ y_1 + y_2\,], \qquad \mathbf{u} - \mathbf{v} = [\,x_1 - x_2,\ \ y_1 - y_2\,].$$

Both are **componentwise**: the first coordinates interact only with each other, and likewise the second. Nothing mixes across slots.

Now notice the *shape* of these two operations, exactly as we did for the norm in §2. Each one is a **binary operator**: it consumes *two* vectors and returns a vector.

$$+\ :\ \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2, \qquad\qquad -\ :\ \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2.$$

Read the left-hand side as "a pair of vectors from $\mathbb{R}^2$ goes in." Contrast this with the norm, which was *unary* — one vector in, one *number* out. Addition and subtraction are binary, and they never leave $\mathbb{R}^2$: two vectors in, one vector out.

Example 7.1 (Both operations on one pair)**.** Let $\mathbf{u} = [2, 1]$ and $\mathbf{v} = [1, 3]$. Then

$$\mathbf{u} + \mathbf{v} = [\,2 + 1,\ 1 + 3\,] = [\,3,\ 4\,], \qquad \mathbf{u} - \mathbf{v} = [\,2 - 1,\ 1 - 3\,] = [\,1,\ -2\,].$$

And since we know how to take a norm: $\|\mathbf{u} + \mathbf{v}\| = \sqrt{9 + 16} = 5$ — the clean vector from §2, arrived at by addition.

7.1. The parallelogram rule

Componentwise arithmetic tells you *what* the sum is. Geometry tells you *where* it is.

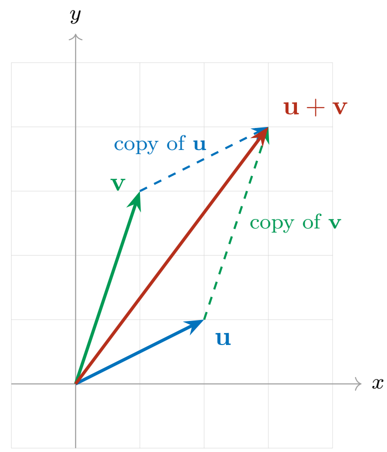
> **Parallelogram rule.** Place $\mathbf{u}$ and $\mathbf{v}$ tail-to-tail at the origin. They span a parallelogram: the two given arrows are one pair of sides, and the copies of them drawn from the opposite corners are the other pair. Then $\mathbf{u}+\mathbf{v}$ is the **diagonal of that parallelogram** running from the common tail to the far corner.

There is a second reading of the same picture, and it is the one to carry around in your head:

> **Head to tail.** Walk along $\mathbf{u}$. From where you land, walk along $\mathbf{v}$. The sum $\mathbf{u} + \mathbf{v}$ is the single arrow from where you started to where you finished.

The two readings agree because the far corner of the parallelogram is reached either way. Notice what the head-to-tail reading quietly does: it draws $\mathbf{v}$ with its tail at the head of $\mathbf{u}$ rather than at the origin. That is legitimate — it is the same vector $\mathbf{v}$, just drawn elsewhere — and §9 is where we make that precise.

In the figure below, the solid arrows are $\mathbf{u}$, $\mathbf{v}$ and their sum from the origin; the dashed arrows are the translated copies that close the parallelogram. Follow the blue solid arrow and then the green dashed one: you arrive at $[3, 4]$, which is the head of the red diagonal.



7.2. The triangle inequality

Take the head-to-tail picture and ignore the parallelogram: what is left is a *triangle*. Its three sides have lengths $\|\mathbf{u}\|$, $\|\mathbf{v}\|$ and $\|\mathbf{u}+\mathbf{v}\|$. And a triangle can never have one side longer than the other two combined — walking straight from start to finish cannot be longer than detouring via a third point.

Triangle inequality. For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$,

$$\|\mathbf{u}+\mathbf{v}\| \;\le\; \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Equality holds in exactly one situation: when $\mathbf{u}$ and $\mathbf{v}$ point in the *same* direction (or one of them is the zero vector). Then the "triangle" has collapsed flat onto a single line, the detour is no detour at all, and the two sides lie end to end along the third.

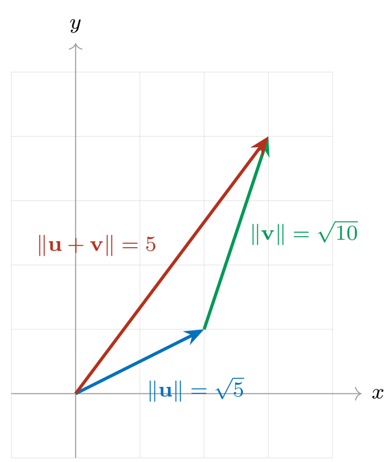
Example 7.2 (Strict inequality)**.** With $\mathbf{u} = [2,1]$ and $\mathbf{v} = [1,3]$ from above, $\mathbf{u}+\mathbf{v} = [3,4]$ and

$$\|\mathbf{u}+\mathbf{v}\| = 5, \qquad \|\mathbf{u}\| + \|\mathbf{v}\| = \sqrt{5} + \sqrt{10} \approx 2.236 + 3.162 = 5.398.$$

Indeed $5 < 5.398$. The two vectors point in genuinely different directions, so the detour costs something.

Example 7.3 (Equality)**.** Let $\mathbf{u} = [1,2]$ and $\mathbf{v} = [2,4]$. Here $\mathbf{v}$ is just $\mathbf{u}$ made twice as long — same direction — so we expect equality. And $\mathbf{u}+\mathbf{v} = [3,6]$, so

$$\|\mathbf{u}+\mathbf{v}\| = \sqrt{9+36} = \sqrt{45} = 3\sqrt{5}, \qquad \|\mathbf{u}\| + \|\mathbf{v}\| = \sqrt{5} + 2\sqrt{5} = 3\sqrt{5}.$$



This inequality is not a curiosity. It is one of the two or three properties that *define* what it means to be a norm at all, and it is the reason distances behave the way our intuition expects.

7.3. Subtraction: the arrow between the heads

Subtraction has its own picture, and it needs no new machinery.

> **Between the heads.** Draw $\mathbf{u}$ and $\mathbf{v}$ from the origin. Then $\mathbf{u} - \mathbf{v}$ is the arrow joining the two heads, pointing *from* the head of $\mathbf{v}$ *to* the head of $\mathbf{u}$.

It is the displacement that carries you from $\mathbf{v}$ to $\mathbf{u}$ — which is why differences of vectors are how we will measure how far apart two vectors are.

That leaves one thing to get right, and it is the step students most often reverse: *which way does the arrowhead go?* Here is a rule that never fails.

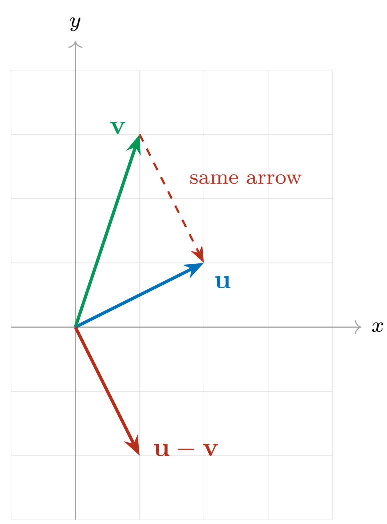
> **Reading a difference as a journey.** Read $\mathbf{v} - \mathbf{u}$ from right to left: *start at* $\mathbf{u}$, *end at* $\mathbf{v}$. So $\mathbf{v} - \mathbf{u}$ points *at* $\mathbf{v}$, and $\mathbf{u} - \mathbf{v}$ points *at* $\mathbf{u}$. The arrowhead always lands on the head of whichever vector is written *first*.

And a numerical check, in case the picture ever betrays you: compute the components and read off their signs. For $\mathbf{v} - \mathbf{u}$, the first coordinate tells you whether the arrow travels right (positive) or left (negative), and the second tells you up (positive) or down (negative).

Example 7.4 (Both directions on one pair)**.** Take $\mathbf{u} = [2, 1]$ and $\mathbf{v} = [1, 3]$ again.

$$\mathbf{u} - \mathbf{v} = [\,1,\ -2\,], \qquad \mathbf{v} - \mathbf{u} = [\,-1,\ 2\,].$$

By the journey rule, $\mathbf{u} - \mathbf{v}$ starts at the head of $\mathbf{v}$, namely $(1, 3)$, and ends at the head of $\mathbf{u}$, namely $(2, 1)$ — one step right and two steps down, matching $[\,1, -2\,]$. The other difference, $\mathbf{v} - \mathbf{u} = [-1, 2]$, is the same segment travelled backwards: one step left and two steps up. The two are opposite arrows on the same line.



The dashed arrow and the solid arrow labelled $\mathbf{u} - \mathbf{v}$ are the *same vector* drawn in two places: same magnitude, same direction, different tail. §9 is where that claim gets its definition.

7.4. Practice: addition and subtraction

For each pair below: **(a)** draw $\mathbf{u}$ and $\mathbf{v}$ on one set of axes, tails at the origin, arrowheads marked; **(b)** compute the indicated sum or difference componentwise; **(c)** draw the resulting vector on the same axes.

P1. $\mathbf{u} = [3, 1]$, $\mathbf{v} = [1, 2]$. Compute $\mathbf{u} + \mathbf{v}$.

P2. $\mathbf{u} = [4, 1]$, $\mathbf{v} = [1, 3]$. Compute $\mathbf{u} - \mathbf{v}$.

P3. $\mathbf{u} = [-2, 3]$, $\mathbf{v} = [3, 1]$. Compute $\mathbf{u} + \mathbf{v}$.

P4. $\mathbf{u} = [2, -2]$, $\mathbf{v} = [-3, -1]$. Compute $\mathbf{u} - \mathbf{v}$.

Optional: compute the norm of each resultant.
Watch the signs in P4: both inputs have a negative component, so $2-(-3) = 5$ and $-2-(-1) = -1$.

Answer key

#	$\mathbf{u}$	$\mathbf{v}$	**Result**	$\lVert$result$\rVert$
P1	$[3, 1]$	$[1, 2]$	$\mathbf{u} + \mathbf{v} = [4, 3]$	5
P2	$[4, 1]$	$[1, 3]$	$\mathbf{u} - \mathbf{v} = [3, -2]$	$\sqrt{13}$
P3	$[-2, 3]$	$[3, 1]$	$\mathbf{u} + \mathbf{v} = [1, 4]$	$\sqrt{17}$
P4	$[2, -2]$	$[-3, -1]$	$\mathbf{u} - \mathbf{v} = [5, -1]$	$\sqrt{26}$

8. Scalar multiplication

Addition and subtraction combine a vector with another vector. The next operation combines a vector with a *scalar* — one of the plain numbers of §4, carrying magnitude but no direction.

Definition 8.1 (Scalar multiplication in $\mathbb{R}^2$)**.** Let $\mathbf{v} = [\,x,\ y\,] \in \mathbb{R}^2$ and let $\alpha \in \mathbb{R}$. Then

$$\alpha\mathbf{v} \;=\; [\,\alpha x,\;\; \alpha y\,].$$

Componentwise again: α multiplies *every* coordinate.
This operation is binary as well, but its two inputs are of *different kinds* — a number and a vector, not two vectors:

$$\mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}^2.$$

A scalar and a vector go in; a vector comes out. That mixed signature is why α is called a *scalar* in the first place: its job is to set the scale of the vector.

Example 8.1 (Scaling the running vector)**.** Let $\mathbf{v} = [\,1,\ 2\,]$, our vector from §1, with $\lVert\mathbf{v}\rVert = \sqrt{5}$.

$$2\mathbf{v} = [\,2,\ 4\,], \qquad \tfrac{1}{2}\mathbf{v} = [\,\tfrac{1}{2},\ 1\,], \qquad -\mathbf{v} = (-1)\mathbf{v} = [\,-1,\ -2\,].$$

Their norms are

$$\lVert 2\mathbf{v}\rVert = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}, \qquad \lVert\tfrac{1}{2}\mathbf{v}\rVert = \sqrt{\tfrac{1}{4}+1} = \tfrac{\sqrt{5}}{2}, \qquad \lVert-\mathbf{v}\rVert = \sqrt{1+4} = \sqrt{5}.$$

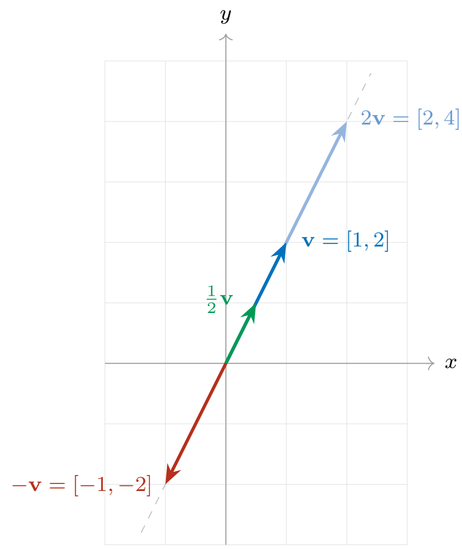
So $2\mathbf{v}$ is twice as long as $\mathbf{v}$, $\frac{1}{2}\mathbf{v}$ is half as long, and $-\mathbf{v}$ is exactly as long as $\mathbf{v}$ but points the opposite way.

Remark. Read off the pattern: scaling by α multiplies the magnitude by $|\alpha|$,

$$\|\alpha\mathbf{v}\| = |\alpha|\,\|\mathbf{v}\|,$$

and the *sign* of α decides the direction — positive keeps it, negative reverses it. That is the same sign-carries-direction rule you saw on the number line in §5, now acting on a vector in the plane. Taking $\alpha = 0$ collapses the vector to $[\,0,\,0\,]$, the **zero vector**, which has no direction at all.

Geometrically, scalar multiplication never turns a vector off its own line: every multiple of $\mathbf{v}$ lies on the single straight line through the origin and the head of $\mathbf{v}$. Scaling slides the arrowhead along that line — stretching it out, pulling it in, or flipping it through the origin — and nothing else.



8.1. What scaling does to the picture

The algebra is one line; the geometry is worth tabulating, because the *sign* and the *size* of α do two independent jobs. The size $|\alpha|$ decides the length; the sign decides the direction.

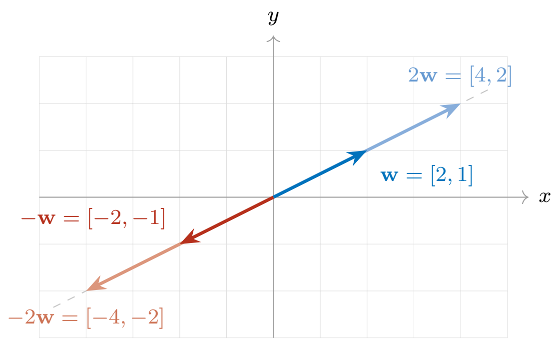
Scalar	**Effect on length**	**Effect on direction**
$\alpha > 1$	stretched	unchanged
$\alpha = 1$	unchanged	unchanged
$0 < \alpha < 1$	shrunk	unchanged
$\alpha = 0$	collapses to the zero vector	destroyed
$-1 < \alpha < 0$	shrunk	reversed
$\alpha = -1$	unchanged	reversed
$\alpha < -1$	stretched	reversed

The figure above showed three of these rows on $\mathbf{v} = [1, 2]$: $\alpha = 2$ stretched it, $\alpha = \frac{1}{2}$ shrank it, $\alpha = -1$ reversed it. The row still missing a picture is the last one — reversing *and* stretching at once — so here it is on a flatter vector, to make the point that none of this is special to $[1, 2]$.

Example 8.2 (Reversing and stretching together)**.** Let $\mathbf{w} = [\,2,\,1\,]$, so $\|\mathbf{w}\| = \sqrt{5}$. Then

$$2\mathbf{w} = [\,4,\,2\,], \qquad -\mathbf{w} = [\,-2,\,-1\,], \qquad -2\mathbf{w} = [\,-4,\,-2\,].$$

Read the last one geometrically: $\alpha = -2$ is negative, so the arrow turns around; and $|\alpha| = 2$, so it comes out twice as long. All four arrows lie on one line through the origin — the two positive multiples on one side, the two negative multiples on the other.



The table also runs backwards, which is often how you will use it: if you can *see* two arrows on a common line through the origin, you can read off the scalar that relates them without computing anything.

Example 8.3 (Reading the scalar off a picture)**.** An arrow is drawn on the same line through the origin as $\mathbf{v} = [1, 2]$. It points the *opposite* way, and it is *three times as long*. Which multiple of $\mathbf{v}$ is it?

Opposite direction forces $\alpha < 0$; three times as long forces $|\alpha| = 3$. Together, $\alpha = -3$, so the arrow is

$$-3\mathbf{v} = [\,-3,\,-6\,].$$

Checking against the algebra: $\|-3\mathbf{v}\| = \sqrt{9+36} = \sqrt{45} = 3\sqrt{5} = 3\|\mathbf{v}\|$, as the picture promised.

Example 8.4 (Scaling a physical vector)**.** Recall from §4 that velocity is a vector quantity. Let $\mathbf{v}$ be the velocity of a car travelling northeast at 30 mph. Then:

- $2\mathbf{v}$ is northeast at 60 mph — same heading, twice the speed;
- $\frac{1}{2}\mathbf{v}$ is northeast at 15 mph — same heading, half the speed;
- $-\mathbf{v}$ is southwest at 30 mph — same speed, reversed heading;
- $0\mathbf{v}$ is the car at rest, with no heading at all.

Scaling changes the speed and, if the scalar is negative, the heading — but it never turns the car onto a new bearing that is neither the original nor its exact reverse.

8.2. Multiplication by -1: mirroring

The case $\alpha = -1$ deserves a name of its own, because it is the one you will use constantly.

> **Mirroring.** $-\mathbf{v} = (-1)\mathbf{v}$ is the **mirror image of v through the origin**: same line, same length, opposite direction. Every coordinate flips sign.

In the figure above, $\mathbf{v} = [1, 2]$ and $-\mathbf{v} = [-1, -2]$ sit on opposite sides of the origin along the same dashed line, at equal distances from it. The origin is the midpoint of the segment joining their two heads.

Remark. Be precise about which mirror. Reflecting *through the origin* — a point — flips *both* coordinates: $[1,2] \mapsto [-1,-2]$. Reflecting *across an axis* — a line — flips only one: across the x-axis, $[1,2] \mapsto [1,-2]$. Multiplication by -1 is the first of these, not the second. Equivalently, it is a half-turn: rotation by π about the origin.

8.3. Subtraction revisited: $\mathbf{u} + (-\mathbf{v})$

In §7.3 we read $\mathbf{u} - \mathbf{v}$ off the picture as the arrow between the two heads. Now that we own the mirror, a second reading opens up — and it lets us reuse the parallelogram rule instead of learning a separate recipe.

Subtraction is nothing but addition of the mirror image:

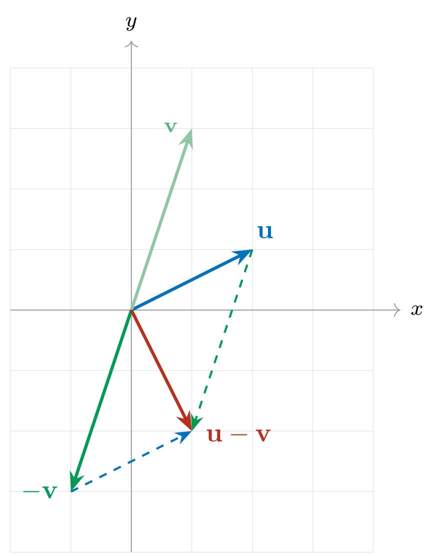
$$\mathbf{u} - \mathbf{v} \;=\; \mathbf{u} + (-\mathbf{v}).$$

So to subtract geometrically: **mirror $\mathbf{v}$ through the origin to get $-\mathbf{v}$, then add $\mathbf{u}$ and $-\mathbf{v}$ by the parallelogram rule of §7.1.** No new geometry at all — one reflection, then the rule you already have.

Example 8.5 (The same difference, computed two ways)**.** Take $\mathbf{u} = [2,1]$ and $\mathbf{v} = [1,3]$ once more. Mirroring gives $-\mathbf{v} = [-1,-3]$, and the parallelogram rule on $\mathbf{u}$ and $-\mathbf{v}$ produces the diagonal

$$\mathbf{u} + (-\mathbf{v}) = [\,2 + (-1),\;\; 1 + (-3)\,] = [\,1,\; -2\,],$$

which is exactly the $\mathbf{u} - \mathbf{v}$ we found in §7.3. Two different pictures, one vector.



The faint green arrow is the original $\mathbf{v}$; the solid green one is its mirror image $-\mathbf{v}$, and the faint dashed line through the origin shows that the two really do lie on a common line. The red diagonal of the parallelogram spanned by $\mathbf{u}$ and $-\mathbf{v}$ is $\mathbf{u} - \mathbf{v}$.

9. When are two vectors equal?

We have twice now drawn "the same vector" in two different places and asked you to believe it. Time to define it.

Definition 9.1 (Equality of vectors)**.** Two vectors are **equal** when they have the same magnitude *and* the same direction. In components, for $\mathbf{u} = [u_1, u_2]$ and $\mathbf{v} = [v_1, v_2]$,

$$\mathbf{u} = \mathbf{v} \qquad \Longleftrightarrow \qquad u_1 = v_1 \ \text{ and } \ u_2 = v_2.$$

Notice what the definition does *not* mention: where the arrow is. A vector is a magnitude together with a direction — a *displacement* — and a displacement does not care where you start it.

Components from any tail: head minus tail

So how do you read the components off an arrow that does not start at the origin? Subtract.

> An arrow with tail $A = (a_1, a_2)$ and head $B = (b_1, b_2)$ is the vector
>
> $$[\, b_1 - a_1, \ \ b_2 - a_2 \,] \qquad — \qquad \textbf{head minus tail}.$$

This is not a new rule; it is the subtraction of §7.3 seen from the other side. Two arrows are the same vector precisely when their head-minus-tail differences agree, no matter how far apart the arrows are drawn. Every vector therefore has infinitely many drawings, called *representatives*, and exactly one of them has its tail at the origin — that one is said to be in **standard position**, and it is the drawing we have been defaulting to since §1.

Example 9.1 (Three drawings, one vector)**.** Each of these arrows is the vector $[\,2,\ 1\,]$:

(i) tail $(0, 0)$, head $(2, 1)$: $\quad [\,2 - 0,\ 1 - 0\,] = [\,2,\ 1\,]$;

(ii) tail $(1, 3)$, head $(3, 4)$: $\quad [\,3 - 1,\ 4 - 3\,] = [\,2,\ 1\,]$;

(iii) tail $(-3, -1)$, head $(-1, 0)$: $\quad [\,-1 - (-3),\ 0 - (-1)\,] = [\,2,\ 1\,]$.

They sit in three different places on the page. They are the same vector: each one says "two steps right, one step up."

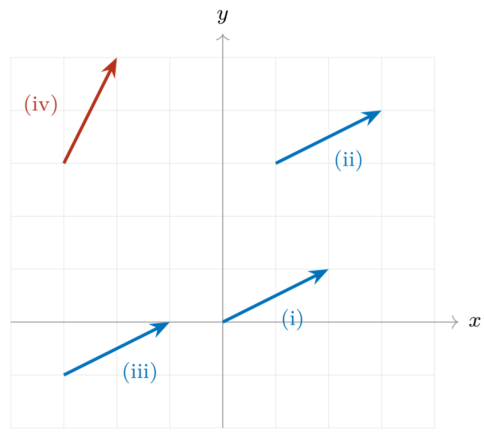
Example 9.2 (Same magnitude is not enough)**.** $[\,1,\ 2\,]$ and $[\,2,\ 1\,]$ both have norm $\sqrt{5}$, so they are equally long — but the first climbs steeply and the second runs flat. Different directions, so they are *different vectors*.

Example 9.3 (Same direction is not enough either)**.** $[\,1,\ 2\,]$ and $[\,2,\ 4\,]$ point exactly the same way — the second is 2 times the first, so by §8 they lie on one line through the origin — but their norms are $\sqrt{5}$ and $2\sqrt{5}$. Different magnitudes, so again *different vectors*.

Example 9.4 (Mirrors are not equal)**.** $[\,1,\ 2\,]$ and $[\,-1,\ -2\,]$ have the same norm $\sqrt{5}$ and lie on the same line, but they point opposite ways. By §8.2 the second is the mirror image of the first, and a mirror image is a *different vector* — unless the vector is $[\,0,\ 0\,]$, which is its own mirror.

Equality needs *both* attributes to match. The last three examples fail on exactly one apiece.

A common confusion. An arrow drawn near the top of the page and an arrow drawn near the bottom *look* like two different objects, and it is tempting to treat them as two different vectors. They are not, so long as head minus tail comes out the same for both. Conversely, two arrows can start at the very same point and still be different vectors. Position on the page is not part of the data; magnitude and direction are.



Arrows (i), (ii) and (iii) are the three representatives of $[\,2,\,1\,]$ from the example above: parallel, equally long, all pointing the same way, and therefore *one and the same vector*. Arrow (iv) is $[\,1,\,2\,]$; it has the same length $\sqrt{5}$ as the others but a different direction, so it is a different vector.